SIMATS ENGINEERING
ASSESSMENT TOOL 1

COURSE CODE: CSA04
COURSE NAME: Operating Systems

COURSE OUTCOME COVERED

CO1: Apply the concepts of Operating System types, structures, and system calls to demonstrate their functions in various computing environments.

QUESTIONS: 5 **Total Marks:100**

Annexure A

SHORT ANSWER TEST

Code of Conduct:

I M. Pranay (192525382)_ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

M. Pranay
Signature of the student:

Annexure A

Short Answer Test

Question 1 (5 Marks)
Suppose a modern Operating System supports multiple users and processes with shared resources like CPU, memory, and I/O devices.
(a) Classify different types of Operating Systems suitable for this scenario and justify your choice.
(b) Draw the structure of an Operating System showing how resources are managed.
(c) Write a sequence of system calls required to create and execute a new process in such an environment. **(BL2)**

Question 2 (5 Marks)
Consider an Operating System managing multiple files and devices for concurrent users.
(a) Categorize system calls used for file and device management.
(b) Illustrate how the OS handles file operations using a layered structure diagram.
(c) Write a pseudo-code or C-based system call sequence to open, read, and close a file. **(BL3)**

Question 3 (5 Marks)
Assume a real-time Operating System used in an embedded system controlling industrial machinery.
(a) Identify the type of OS and explain why it is suitable for real-time applications.
(b) Draw a diagram representing the interaction between hardware, kernel, and user processes.
(c) List and explain system calls required for process synchronization in this system. **(BL2)**

Question 4 (5 Marks)
A distributed computing system consists of multiple interconnected computers sharing resources over a network.
(a) Classify the type of Operating System used and compare it with centralized OS.
(b) Draw the architecture of a distributed OS showing communication between nodes.
(c) Write the sequence of system calls involved in inter-process communication (IPC). **(BL3)**

Question 5 (5 Marks)
Consider a scenario where an Operating System must manage process scheduling, memory allocation, and device communication efficiently.
(a) Explain the major functions of an OS in handling these tasks.
(b) Draw a block diagram representing OS components and their interactions.
(c) Write system call examples for memory allocation and device management. **(BL3)**

RUBRICS FOR CALCULATION

Criteria	Weightage (Marks)	Level 4 – Exemplary (Full Marks)	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Conceptual Understanding	35 Marks	Demonstrates clear, complete, and in-depth understanding of OS concepts (types, structure, system calls)	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answers	25 Marks	All answers are correct, precise, and relevant	Mostly correct with minor errors	Partially correct with noticeable errors	Mostly incorrect or irrelevant
Application of Knowledge	15 Marks	Effectively applies concepts to scenarios/questions	Applies with minor mistakes	Limited or partial application	No meaningful application
Completeness of Response	15 Marks	All parts of each question are fully answered	Most parts covered with small omissions	Some parts missing	Incomplete answers
Clarity & Presentation	10 Marks	Clear, well-structured, and neatly presented answers	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

Assessment Tool-1

Short Answer Test

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Reg No :- 192525382

Course Code :- UBA0408

Course Name :- Operating system.

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1) a) Types of operating system suitable for multiple users and processes.

For a modern OS supporting multiple users, processes, CPU, memory and I/O devices.

1) Multi-user operating system :-
   → Provides security and resource sharing
   → Linux, UNIX.

2) multiprogramming operating system :-
   → keeps multiple programs in memory at the same time.
   → CPU switches b/w processes to improve CPU utilization.

3) Multi tasking Operating System:-

→ Allows multiple Processes to execute Concurrently.

→ windows, linux.

4) Time sharing Operating system:-

→ Provides quick response to interfactive users

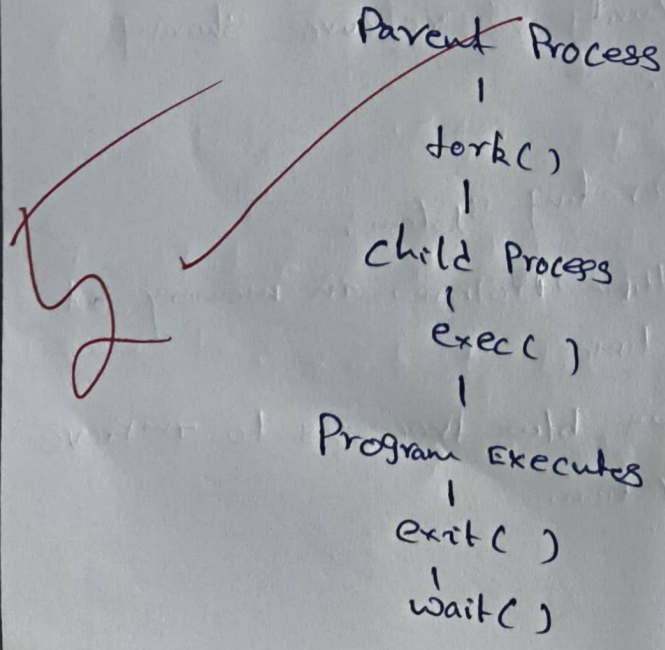
5) MultiProcessing Operating System:-

→ Supports multiple CPUs.

b) structure of an Operating system showing resourse management.

The OS manages CPU, manory, tiles and I/O devices and Provides services to users and applications.

C) Process creation sequence:-



2) System Call Categories:-

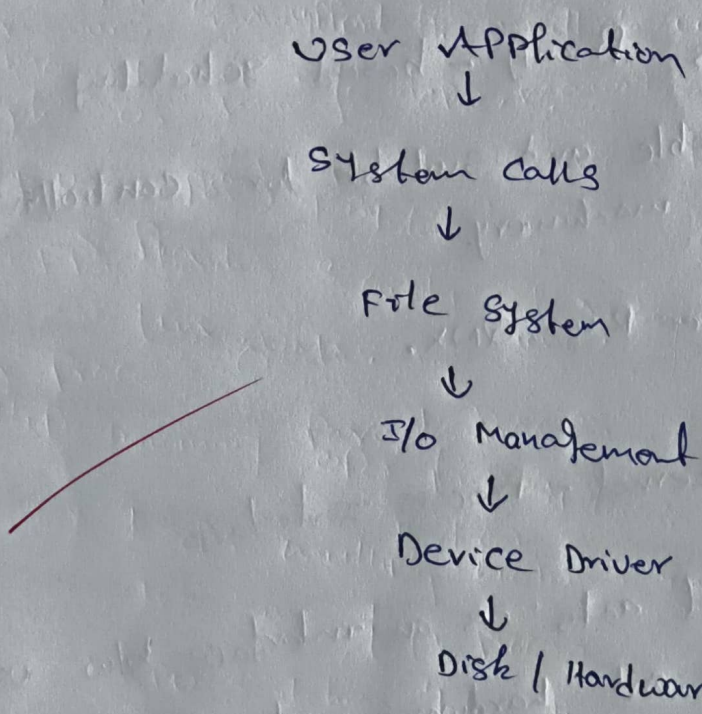
File management:-

→ open() - opens a file.
→ read() - reads data
→ write() - writes data
→ close() - closes a file.

Device management:-

→ open() - opens a device
→ read()/write() - transfers data.
→ ioctl() - controls device operations.
→ close() - releases devices.

b) layered File structure.



c) File system call sequence:-

open() → read() → close()

3) (a) Type of OS:-

The suitable OS is a Real Time Operating System (RTOS).

→ RTOS gives a fast and predictable response to events.

→ It ensures that important tasks are completed within a specified time.

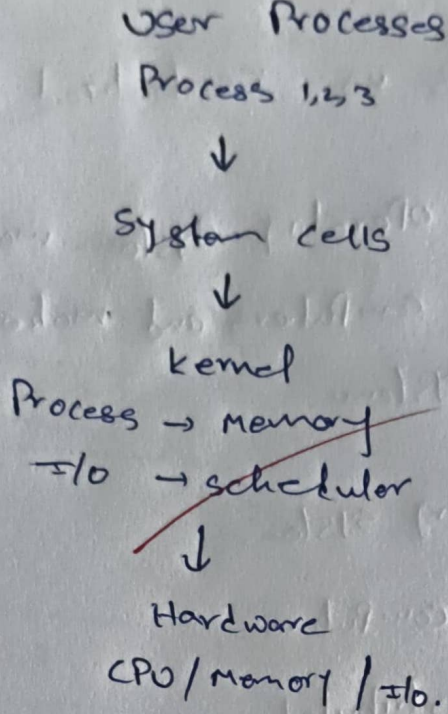
→ It Provides Priority - based scheduling

→ It is reliable and suitable for controlling industrial machinery.

Ex:- Free RTOS, QNX.

b) Hardware - kernel - Process Interface

The kernel acts as an interface b/w user and Processes and hardware. It manages CPU, memory and I/O resources.



c) System calls for Process synchronization:-

1) Semaphore operations - wait() and signal()
control access to shared resources.

2) mutex lock/unlock - lock() and unlock()
Provides mutual exclusion.

3) Condition variables:- allow Processes to wait for a particular condition.

4) Message Passing:- send() and receive() allow Processes to communicate safely.

5) wait/signal:- Processes can wait for or signal the completion of another Process.

4) a) Distributed OS vs Centralized OS.

→ A Distributed operating system manages multiple interconnected computers and makes them work like a single system.

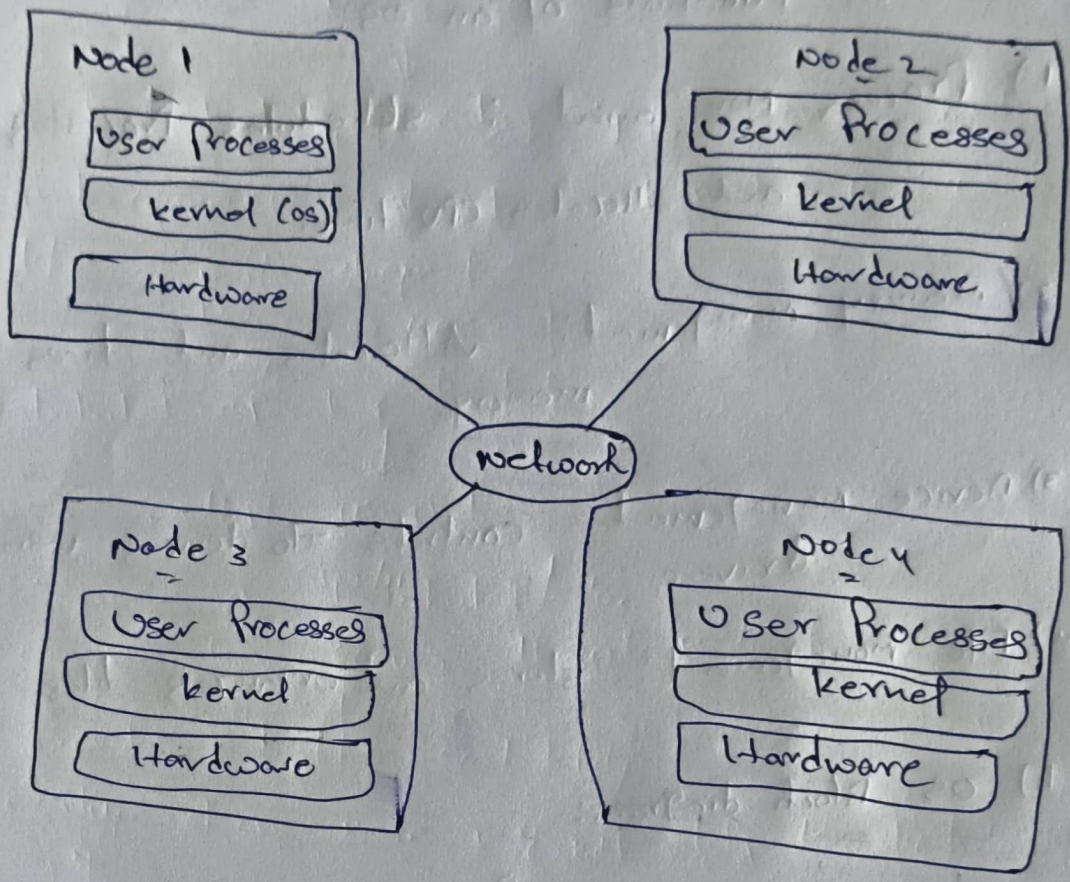
Distributed operating system:-

→ Uses multiple computers.
→ Resources are distributed.
→ Communication occurs through network.
→ Provides better scalability.
→ Failure of one node may not stop the whole system.

Centralized operating system:-

→ Uses mainly one central computer
→ Resources are centralized.
→ local communication.
→ limited scalability.
→ central failure can affect the system.

b) Distributed OS Architecture:-



C) Inter Process communication system calls:-

→ Pipe() → Creates communication b/w Processes.
→ send() → sends data.
→ receive → Create receives data.
→ shmget() → Creates shared memory.
→ mmap() → maps shared memory.

5) a) Major Functions of an OS:-

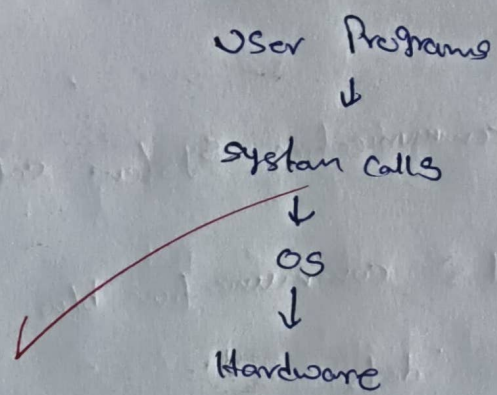
1) Process management :- schedules Processes and allocates CPU time.

2) Memory management :- Allocates and frees memory.

3) Device management :- Controls I/O devices using drivers

4) File management :- stores and organizes files.

b) OS Block diagram :-



c) System Calls :-

Memory :- mmap( ), munmap( )

Device :- Open( ), read( ), write( ), ioctl( ).